Edward Woodruff

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ACADEMICS

Aerospace Engineering (B.S.) | Math (minor) **May 2025**
University of Central Florida *GPA 3.73*

- Undergraduate Research: high heat-flux, pulsed, evaporative spray cooling.
- Software + Controls Lead, SABR (Small-Scale Air Breathing Rotating Detonation Engine)

EXPERIENCE

Tesla | Manufacturing Test Engineer **Austin, TX**
Drive Inverter + Energy *June 2025 – Present*

- Co-developed a Python firmware flashing framework and authored 3 production implementations with custom drivers for JTAG, SPI, I2C, and CAN protocols (XDS200, Dediprog, Renesas, J-Link, Aardvark, PCAN), deploying across 10+ testers and reducing cycle time by 40% while improving yield by 5% via Python's multiprocessing module.
- Maintained and contributed to 15+ repositories across Python, LabVIEW/TestStand, Beckhoff, and Go - independently driving software improvements across 50+ testers in 8 production facilities, spanning bug fixes, feature development, sustaining updates, code reviews, and on-call support across the full software lifecycle.
- Owned the full tester deployment for 3 projects, leading Factory Acceptance Testing (FAT) and Site Acceptance Testing (SAT) to validate GR&R and CPK requirements, optimize operator usability, and resolve issues prior to handoff, with ongoing software sustaining support post-deployment.

Tesla | Abuse Test Systems Engineer **Fremont, CA**
Test Systems Engineering Intern *May – Sep. 2024*

- Created a novel MATLAB script to streamline CAN trace post-processing, automating the conversion from multiplexed hexadecimal data to digestible plots and reducing engineering time by ~1.8 hours per test
- Owned all controls programming (LabVIEW) for a custom Vehicle Crusher test rig. This system consisted of a load cell, limit switch, string pot, VFD, encoder, thermocouples, and a series of E-Stop switches, ensuring safe operation and precise execution of crush-to-propagation thermal event tests.
- Authored PLC & HMI code to log and monitor test chamber airspeed, temperature, and test execution rate to deepen understanding of test characteristics, measure throughput, and track cell propagations.

Tesla | Battery Abuse Test Engineer **Palo Alto, CA**
Thermal Test Engineering Intern *Sep. 2023 – May 2024*

- Developed a novel MATLAB script to analyze experiment videos to calculate cell propagation rate during thermal event tests using image and acoustic analysis, saving engineers 3h per test in processing time.
- Authored a custom LabVIEW script to conduct flammability tests by automating gas flow to adhere to any test requirements or flow conditions including constant, PWM (pulse width modulation), and vector flow
- Designed and conducted coupon, module, pack, and vehicle-level forced-propagating thermal runaway tests safely and in a manner faithful to the design intent / failure mode.

SpaceX | PCB Supplier Development Engineer **Redmond, WA**
Starlink Engineering Intern *May – Aug. 2023*

- Developed, tested, and integrated a new PCB stud mounting technique which generated $6.3m in savings, reduced satellite weight by 1.5kg, freed up production floor space by 15%, and cut build time by 80 minutes per board on a high-volume assembly.
- Engineered and developed multiple Design of Experiments (DOE) to ensure the optimal selection of variables including global and Via in Pad (VIP) stitching, solder pad size, stencil aperture design, and solder paste volume.
- Analyzed failure modes associated with PCB scrap rate and qualified rework procedures to increase global PCB yield by 1.7% and reduce spend by $1.3m in 2024 without introducing additional reliability concerns.

TECHNICAL SKILLS

- **Skills:** Embedded Systems, Multiprocessing, Test-Automation, CI/CD, AI-Assisted Development (Claude Code, Copilot),
- **Programming**: Python, LabVIEW, NI-Teststand, MySQL, MATLAB, Siemens TIA Portal (PLC / HMI), Beckhoff, Go
- **Software**: VScode, GitHub, SolidWorks, Catia V5/3DX, AutoCAD, Confluence, Jira
- **Interfaces:** CAN, JTAG, SPI, I2C, Serial, USB, Ethernet